cloudtorch / develop — Step 5a debug: relaxing spatial regularisation

2026-08-24 — fits from `step_5a_reg_test.ipynb` (cell 20) were unsatisfactory. Hypothesis tested: the explicit **spatial** smoothness was over-constraining the cloud maps. Action: relax it $100\times$ and re-run to recover a good data fit before re-tightening.

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1. WHAT WAS RUN

The step-5a notebook config reproduced verbatim (given fixed background, `n_bkg`=0; `CROP`=64, `N_T`=8; 10^4 cosine-annealed Adam iters), changing *only* the spatial weight w^{xy}: $10^{-3} \to 10^{-5}$ across all four cloud groups (`S`,`dtau`,`vlos`,`dv`). Temporal w^t and the soft hinges were left untouched.

2. RESULT: ESSENTIALLY NO CHANGE

run	$\langle\chi^2\rangle$	RMS resid	v_1 [min/med/max]	v_2 [min/med/max]
$w^{xy} = 10^{-3}$ (baseline)	0.460	5.36%	$-79.8/-24.4/-22.7$	$11.1/22.1/23.1$
$w^{xy} = 10^{-5}$ ($100\times$)	0.416	5.09%	$-76.9/-24.9/-21.4$	$10.9/22.2/22.9$

χ^2 fell only because the penalty itself shrank; the *data* fit did not improve. The parameter maps are unchanged: `vlos1` uniformly blue, `vlos2` uniformly red, neither crossing zero, both pinned within ~ 2 km/s of the ∓ 20 initialisation. The pixel spectra still miss the blue-wing asymmetry (pix 2575, 1608).

3. WHY RELAXING REG COULD NOT HELP (THE FINDING)

There was no *binding* spatial regularisation to relax. In both runs the total reg is only $\sim 2.5\%$ of the loss (0.011 vs data 0.416) and every hinge is inactive (velocities sit at ∓ 22, far inside the ± 120 range limit and below the $\Delta v = 45$ cap). The fit is therefore **already effectively unregularised** — so "loosen reg, then tighten" cannot get traction. The mediocre fit and the flat velocities are intrinsic to the *setup*, not the penalties.

4. ROOT CAUSE OF THE VELOCITY COLLAPSE

The two clouds collapse to a symmetric ∓ 22 km/s pair because (i) the **given background already owns the line wings**, where the Doppler signal lives, so the clouds get almost no velocity leverage; and (ii) the **symmetric** ∓ 20 **init** is a flat spot the data gradient never leaves — line *depth* (via $S, \Delta\tau$) dominates, velocity is second-order. So $S/\Delta\tau$ carry all the fitting (real fibril structure) while the Doppler DoF stay dead.

5. LOSS CURVE

Converged by $\sim$1–2k iters ($8.16 \to \sim 0.42$), then flat for $\sim$8k — the extra 90% of the 711 s run buys nothing (a degeneracy floor, not the noise floor: RMS 5% is well above photon noise). Periodic order-of-magnitude spikes coincide with reg spikes: the autograd-smoothness double-backward occasionally emits a large gradient, Adam overshoots, then recovers.

6. NEXT LEVER (NOT A REG KNOB)

Attack the **velocity init / wing access**, one change at a time: **(1)** asymmetric `vlos` init (e.g. -5 / $+35$) or a per-pixel line-bisector warm-start, to escape the symmetric trap — cheapest test; **(2)** if needed, let the clouds reach the wings the background currently claims. Only after the velocity field moves should the smoothness/hinge weights be revisited.

Notebook `cell-3` now carries `W_XY=1e-5` behind a single tightening knob; the `../figs/step_5a_pinn_*.pdf` fit-image and parameter-map figures are refreshed.